



D2 Change Report

Primary: Emma Fuller

Secondary: Anexa Bijoy

Tertiary: Leo Durnion, Wasif Mustafa, Samuel Knight, Hamza Ahmad

D2a - Processes, Tools, & Convention

Changes to deliverables

In order to keep track of changes to previous deliverables we made sure to keep an original copy of all of the documents so that we could refer back to them as needed. We then followed the steps outlined below when changing out documentation inspired by [1].

1- **Request** - First the changes are made to the deliverables by using the suggestion mode on Google Docs. This ensures that the original document is not changed until the change has been reviewed. The changes were made either due to the new brief for assessment 2 or as we felt like we could improve on the documentation provided by the previous team.

2- **Review** - We had another member of our team who is knowledgeable in that area to check that the change is needed. They also made sure that the relevant team members are aware of the possible impacts of the change; for example, a new requirement could mean another test is needed. They need to ensure that the change will add value to the project and not cause unnecessary work for other members of the team.

3- **Implementation** - Once the change is reviewed, then it can be added into the document by accepting the change.

4 - **Closure** - As part of this step, the change will be added to the change report, so we can keep track of all of the changes that have been made and with the justifications.

We used the Kanban chart on Github to assign tasks to team members so that it is clear who is working on which document to avoid conflicts. This allows us to set internal deadlines so that any delays due to dependencies are limited.

By using Google Docs, it also means that we have access to the edit history and version control so we can convert back to previous versions if we feel like we need to as a team.

Changes to code

We have once again kept an original version of the code to aid our debugging process.

We will discuss any changes that need to happen to the original code before we start adding new functionality or fixes identified during the testing phase and these will be added to the Kanban. When we do decide to modify the code, we will assign tasks on GitHub to ensure the most effective use of our time and resources.

All changes and additions to the code will be completed on a separate branch then the pull request will be reviewed by another member of the implementation team to ensure that they were as expected and that the code is efficient and formatted correctly.

Changes in the code will also be marked with a `'// CHANGED: '` tag so that it is clear to all members of the team where our additions are and will be added to the change report.

Changes to website

The website is on a smaller scale than the game repository, so we can rely more heavily on version history to access the changes that have been made. The changes will still be added to the change report alongside a detailed justification so we can record why certain changes were made.

D2b - Justification of Changes

Requirements

Old Version	New Version
https://whales-eng1.github.io/WHALES-Website-Assessment-2/documents/assessment1/Req1.pdf	https://whales-eng1.github.io/WHALES-Website-Assessment-2/documents/assessment2/d9%20Requirements.pdf

Change	Justification
Added in a paragraph about the research that has been done on how to format the requirements	This was specified in the assessment document but the previous team had not included it.
Removed requirement NFR_FILE_SIZE	We spoke to the customer and they had not specified a file size so this requirement was not given by the customer so is therefore not needed.
Added in 3 constraint requirements	The previous team had not addressed any constraint requirements and we felt like they were important to recognise in our development process.
Added in the new requirements for assessment 2	Building upon the original project, adding requirements for the product brief for assessment 2.
Added in UR_MUSIC, FR_MUSIC and FR_MUSIC_MUTE	This was a requirement we had asked about in our customer meeting and had been implemented in the game but not included in the requirements.
Added in FR_TIMER_END	This is a functional requirement based on a user requirement which the previous group had stated.
Added in FR_WIN, UR_WIN_CONDITION	This condition needs to be met in order for the game to be playable by our audience.
FR_CONTROLS, FR_RESUME	This requirement has already been met but needs to be added so we can reference it whilst testing.

Architecture

Old Version	New Version
https://whales-eng1.github.io/WHALES-Website-Assessment-2/documents/assessment1/Arch1.pdf	https://whales-eng1.github.io/WHALES-Website-Assessment-2/documents/assessment2/d8%20Architecture.pdf

Change	Justification
Added new screen - AchievementScreen	A new screen that was not initially implemented. We added it to the PlantUML diagram that was already created by Team 1 following the same structure they used. Allows the user to view the achievements they've earned so far in the run, as well as other achievements they may earn.
Added new screen - UsernameScreen	A new screen that was not initially implemented. We added it to the PlantUML diagram that was already created by Team 1 following the same structure they used. Allows the user to input a username of maximum four characters.
Added all separate room screens: BaseLibraryScreen, BurtonScreen, FairHurstScreen, LibraryBasementScreen, MorrellScreen, RoomScreen	We branched off the individual rooms that make up the map into the PlantUML diagram, as they are all small we kept the information to a minimum to prevent the diagram from becoming overcomplicated.
Add new entity - Dean	We added the Dean to the game which builds on the structure developed by the previous team and works seamlessly into the PlantUML diagram.

Method Selection and Planning

Old Version	New Version
https://whales-eng1.github.io/WHALES-Website-Assessment-2/documents/assessment1/Plan1.pdf	https://whales-eng1.github.io/WHALES-Website-Assessment-2/documents/assessment2/d7%20Method%20Selection%20&%20Planning.pdf

Change	Justification
Added a paragraph on our adjusted methodology	Our agile methodology approach slightly differed to the previous team in terms of the days allocated for meetings.
Added our form of communication	We were familiar with discord already, and have set up different channels based on the new deliverables for this assessment, so continuing with this form of communication felt the most reasonable for us.
Added our method for planning	Continuing using Github projects instead of Trello and PlantUML (which were used by the previous team for Kanban boards and Gantt charts), made sense as we were already familiar with using GitHub Projects from Assessment 1. We used GitHub projects to create our Kanban boards which

Change	Justification
	differed to the software of the previous team. However, we continued the use of plantUML to create the Gantt charts.
Added Paint.net and Piskell	Team 1 used photoshop which is a paid software which we do not have access to so we will be using Paint.net and Piskell to create our assets.
Added in our team structure and justification	We used a slightly different system to the previous group as we assigned primaries and secondaries to each deliverable and added justification and review of how it worked for us in assessment 1.
Added in justification for choice of methodology	There was no justification in the document so we added in reasons why agile was used instead of other methodologies like waterfall.
Added plan for each period of Assessment 2 including Gantt charts	We made a new plan to complete the new assessment criteria and therefore created new Gantt charts to show how our team has worked in weeks 7-11 to complete this assessment.

Risk Assessment and Mitigation

Old Version	New Version
https://whales-eng1.github.io/WHALES-Website-Assessment-2/documents/assessment1/Risk1.pdf	https://whales-eng1.github.io/WHALES-Website-Assessment-2/documents/assessment2/d11%20Risk%20Assessment.pdf

Change	Justification
Owners of the risk	Since we are a new team, we assigned each risk to our own team member who will now take ownership of those risks.
Remove "R4 - Assets were all made by the group not sourced which took a while"	As a team, we did not consider this to be a risk. Most assets for the game were already made for us during Assessment 1, and the creation of assets is simply a necessary part of the process of developing the game.
Remove "R10 - Lack of early user feedback before handover"	Due to the changed requirements for assessment 2, user feedback is now a required aspect of the project, so lack of it is no longer a risk.
Remove "R12 - Lack of prior experience with LibGDX"	Our team does all have prior experience with LibGDX via Assessment 1, so this does not apply to our team and is no longer a risk.
Added justification for the system used	The original document did not justify their methodology for handling risks.

References

[1]K. Luong, "Complete Guide to Change Management in Software Development," *Ominext.com*, 2024.

<https://www.ominext.com/en/blog/change-management-in-software-development>